

# A Rank-One Counterexample to Random-Order Contention Resolution under Setwise Negative Correlation

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## Abstract

We give a five-element counterexample to the following rank-one random-order contention-resolution question: does the setwise negative-correlation condition

$$\mathbb{P}\left[\bigwedge_{e \in S} (Z_e = b)\right] \leq \prod_{e \in S} \mathbb{P}[Z_e = b] \quad (S \subseteq E, b \in \{0, 1\})$$

together with  $\sum_e \mathbb{P}[Z_e = 1] \leq 1$  guarantee a  $(1 - 1/e)$ -selectable random-order scheme? The answer is no. The example satisfies the displayed inequalities for both all-ones and all-zeros events and has total marginal mass exactly one, but no random-order online selector can be  $(1 - 1/e)$ -selectable. More quantitatively, its best common selectability is at most

$$\frac{2290330421101}{3635200000000} = 0.630042479396 \dots < 1 - \frac{1}{e}.$$

Thus the result is a full negative solution to that formulation of the open problem, not merely partial progress. The only qualification is that the result refutes the property above as a sufficient hypothesis by itself; it does not address narrower subclasses of negatively correlated distributions arising from a specified rounding procedure.

## 1 Introduction and status of the result

A rank-one matroid on a finite ground set  $E$  permits selecting at most one element. Let  $Z = (Z_e)_{e \in E} \in \{0, 1\}^E$  denote the activation vector, and write

$$z_e := \mathbb{P}[Z_e = 1].$$

We consider the negative-correlation property

$$\mathbb{P}\left[\bigwedge_{e \in S} (Z_e = b)\right] \leq \prod_{e \in S} \mathbb{P}[Z_e = b] \quad \text{for all } S \subseteq E \text{ and } b \in \{0, 1\}. \quad (1)$$

This is the property denoted as property (2) in the discussion of contention resolution under negative correlation in Brubach, Grammel, Ma, MacRury, and Srinivasan [1]. They show that a  $(1 - 1/e)$  guarantee is possible when a suitable order can be chosen, and ask whether the same guarantee is possible in the random-order model under (1) alone.

The theorem below answers this question negatively. In the model used here, a uniformly random permutation of all elements is revealed online. On arrival of an element, the selector observes its identity and whether it is active. If the element is active, the selector may irrevocably select it; after

selecting one element it stops. The selector may know the full activation distribution and may be randomized. This is a generous model for an impossibility result: revealing inactive arrivals and the full distribution can only help the selector.

A selector is called  $c$ -selectable if

$$\mathbb{P}[e \text{ is selected}] \geq cz_e \quad (e \in E).$$

Equivalently, for  $z_e > 0$ , the conditional probability of selecting  $e$  given  $Z_e = 1$  is at least  $c$ .

**Theorem 1.** *There is a distribution on five activation indicators  $Z_0, Z_1, Z_2, Z_3, Z_4$  satisfying (1) and*

$$\sum_{i=0}^4 \mathbb{P}[Z_i = 1] = 1$$

*for which no random-order rank-one selector is  $(1 - 1/e)$ -selectable. Indeed, the best common selectability on this instance is at most*

$$\frac{2290330421101}{3635200000000} < 0.630043 < 1 - \frac{1}{e}.$$

## 2 The five-element distribution

Let

$$a = \frac{1}{100}, \quad p = \frac{1-a}{4} = \frac{99}{400}.$$

Element 0 is special. Its marginal is  $z_0 = a$ , while the four ordinary elements have marginal

$$z_i = p \quad (1 \leq i \leq 4).$$

Thus

$$z_0 + z_1 + z_2 + z_3 + z_4 = a + 4p = 1.$$

The activation distribution is defined as follows.

- (i) With probability  $a$ ,  $Z_0 = 1$ . Conditional on  $Z_0 = 1$ , the ordinary vector  $(Z_1, Z_2, Z_3, Z_4)$  is either all zero with probability  $a$ , or has exactly one active ordinary element  $i$  with probability  $p$  for each  $i \in \{1, 2, 3, 4\}$ . These conditional probabilities sum to  $a + 4p = 1$ .
- (ii) With probability  $1 - a$ ,  $Z_0 = 0$ . Conditional on  $Z_0 = 0$ , the variables  $Z_1, Z_2, Z_3, Z_4$  are independent Bernoulli( $p$ ) variables.

For each ordinary element  $i$ ,

$$\mathbb{P}[Z_i = 1] = ap + (1-a)p = p,$$

as claimed.

**Lemma 1.** *The distribution above satisfies (1) for every subset  $S \subseteq \{0, 1, 2, 3, 4\}$  and both  $b \in \{0, 1\}$ .*

*Proof.* The empty set is immediate, so assume  $S \neq \emptyset$ .

First take  $b = 1$ . If  $0 \in S$  and  $S$  contains  $k$  ordinary elements, then

$$\mathbb{P}[Z_0 = 1 \text{ and all those } k \text{ ordinary elements are active}] = \begin{cases} a, & k = 0, \\ ap, & k = 1, \\ 0, & k \geq 2. \end{cases}$$

This is at most  $ap^k = \prod_{e \in S} z_e$ . If  $0 \notin S$  and  $|S| = k$ , then the probability that all elements of  $S$  are active is  $p$  for  $k = 1$  and  $(1 - a)p^k$  for  $k \geq 2$ , which is at most  $p^k = \prod_{e \in S} z_e$ .

Now take  $b = 0$ . If  $0 \in S$  and  $S$  contains  $k$  ordinary elements, then, conditional on  $Z_0 = 0$ , the ordinary variables are independent Bernoulli( $p$ ), and hence

$$\mathbb{P}[Z_0 = 0 \text{ and those } k \text{ ordinary elements are inactive}] = (1 - a)(1 - p)^k = \prod_{e \in S} (1 - z_e).$$

If  $0 \notin S$  and  $|S| = k$ , then conditional on  $Z_0 = 1$  all elements of  $S$  are inactive with probability

$$a + (4 - k)p = 1 - kp,$$

because either no ordinary element is active or the unique active ordinary element lies outside  $S$ . Therefore

$$\mathbb{P}[\text{all elements of } S \text{ are inactive}] = a(1 - kp) + (1 - a)(1 - p)^k.$$

By Bernoulli's inequality,  $(1 - p)^k \geq 1 - kp$  for  $p \in [0, 1]$ , so this is at most

$$a(1 - p)^k + (1 - a)(1 - p)^k = (1 - p)^k = \prod_{e \in S} (1 - z_e).$$

This proves the claim. □

### 3 A weighted obstruction

We prove impossibility by assigning weights

$$w_0 = 43, \quad w_i = 1 \quad (1 \leq i \leq 4).$$

For any selector, let  $W$  be the selected weight, with  $W = 0$  if nothing is selected. If the selector is  $c$ -selectable, then

$$\mathbb{E}[W] = \sum_{i=0}^4 w_i \mathbb{P}[i \text{ is selected}] \geq c \sum_{i=0}^4 w_i z_i = c(43a + 4p) = c \cdot \frac{71}{50}. \quad (2)$$

In particular, a  $(1 - 1/e)$ -selectable selector would have

$$\mathbb{E}[W] > \frac{79}{125} \cdot \frac{71}{50} = \frac{5609}{6250} = 0.89744,$$

because  $e > 125/46$ , and hence  $1 - 1/e > 79/125$ .

It remains to prove that even the optimal weighted random-order selector has expected selected weight less than this value.

**Proposition 1.** *For the distribution above, every random-order online selector that selects at most one active element has*

$$\mathbb{E}[W] \leq \frac{2290330421101}{2560000000000} = 0.894660320742578125 \dots < \frac{179}{200}.$$

*Proof.* We compute the optimal value by dynamic programming. Because the ordinary elements are exchangeable and have equal weight, the value after any history depends only on:

- $r$ , the number of ordinary active arrivals already rejected;

- $s$ , the number of ordinary inactive arrivals already seen;
- the status of the special element.

Let

$$m = 4 - r - s$$

be the number of unseen ordinary elements. The special element has one of three statuses:

- $I$ : the special element has already arrived inactive;
- $A$ : the special element has already arrived active and was rejected;
- $U$ : the special element has not yet arrived.

Let  $V_I(r, s)$ ,  $V_A(r, s)$ , and  $V_U(r, s)$  be the optimal future expected selected weight from these states, conditional on no element having yet been selected.

If the special element has arrived inactive, then we know  $Z_0 = 0$ , and the remaining ordinary elements are independent Bernoulli( $p$ ). Thus  $V_I(r, s) = 0$  for  $m = 0$ , and for  $m > 0$ ,

$$V_I(r, s) = p \max\{1, V_I(r + 1, s)\} + (1 - p)V_I(r, s + 1). \quad (3)$$

If the special element has arrived active and was rejected, then we know  $Z_0 = 1$ . Conditional on  $Z_0 = 1$ , at most one ordinary element is active. Therefore  $V_A(r, s) = 0$  when  $m = 0$  or  $r \geq 1$ . When  $r = 0$  and  $m = 4 - s > 0$ , the posterior total mass of the possibilities for the unseen ordinary elements is  $a + mp$ : either no ordinary element is active, with mass  $a$ , or one of the  $m$  unseen ordinary elements is active, with mass  $p$  each. Hence the next ordinary element is active with probability  $p/(a + mp)$ , and

$$V_A(0, s) = \frac{p}{a + mp} \max\{1, V_A(1, s)\} + \left(1 - \frac{p}{a + mp}\right) V_A(0, s + 1). \quad (4)$$

Now suppose the special element is unseen. For an ordinary history with parameters  $(r, s)$ , define unnormalised posterior likelihoods

$$\alpha_{r,s} = \begin{cases} a(a + mp), & r = 0, \\ ap, & r = 1, \\ 0, & r \geq 2, \end{cases} \quad \beta_{r,s} = (1 - a)p^r(1 - p)^s, \quad D_{r,s} = \alpha_{r,s} + \beta_{r,s}.$$

Here  $\alpha_{r,s}$  is the likelihood of the observed ordinary history together with  $Z_0 = 1$ , while  $\beta_{r,s}$  is the corresponding likelihood together with  $Z_0 = 0$ . Therefore, if the special element is the next arrival, its conditional probability of being active is

$$\theta_{r,s} = \frac{\alpha_{r,s}}{D_{r,s}}.$$

If instead a uniformly random unseen ordinary element is the next arrival, its conditional probability of being active is

$$q_{r,s} = \frac{\mathbf{1}_{\{r=0\}}ap + (1 - a)p^{r+1}(1 - p)^s}{D_{r,s}}.$$

Since the next element among the  $m$  unseen ordinary elements and the one unseen special element is uniform, we have

$$V_U(r, s) = \frac{1}{m+1} [\theta_{r,s} \max\{43, V_A(r, s)\} + (1 - \theta_{r,s}) V_I(r, s)] \quad (5)$$

$$+ \frac{m}{m+1} [q_{r,s} \max\{1, V_U(r+1, s)\} + (1 - q_{r,s}) V_U(r, s+1)], \quad (6)$$

where the second line is omitted when  $m = 0$ .

The system (3), (4), (5) is acyclic: every transition either changes the special status from  $U$  to  $A$  or  $I$ , or reduces  $m$ . Hence backward induction gives the exact optimal value. Substituting

$$a = \frac{1}{100}, \quad p = \frac{99}{400}$$

and evaluating with exact rational arithmetic gives

$$V_U(0, 0) = \frac{2290330421101}{2560000000000} = 0.894660320742578125 \dots < \frac{179}{200}.$$

Appendix A gives a self-contained exact-rational verifier. It also computes the same optimal value directly from the full activation distribution, without using the reduced symmetry states above. This proves the proposition.  $\square$

## 4 Proof of the main theorem

Assume for contradiction that a  $(1 - 1/e)$ -selectable random-order selector exists for the five-element distribution. By (2), using the weights  $w_0 = 43$  and  $w_i = 1$  for  $1 \leq i \leq 4$ , its expected selected weight must be strictly larger than

$$\frac{5609}{6250} = 0.89744.$$

On the other hand, Proposition 1 says that every random-order online selector has expected selected weight at most

$$\frac{2290330421101}{2560000000000} < \frac{179}{200} = 0.895.$$

Since

$$\frac{179}{200} < \frac{5609}{6250},$$

this is impossible. Thus no  $(1 - 1/e)$ -selectable random-order rank-one selector exists under (1) alone.

Equivalently, for this instance every selector has common selectability at most

$$\frac{V_U(0, 0)}{43a + 4p} = \frac{2290330421101}{3635200000000} = 0.6300424793961817 \dots < 1 - \frac{1}{e}.$$

This proves Theorem 1.

*Remark 1* (Scope of the counterexample). The counterexample refutes the statement that the setwise negative-correlation property (1), together with  $\sum_e z_e \leq 1$ , is sufficient for a  $(1 - 1/e)$ -selectable random-order CRS in rank one. It does not contradict ordered contention-resolution results, and it does not by itself rule out a  $(1 - 1/e)$  random-order guarantee for some narrower structured class of activation distributions satisfying (1).

## A Exact arithmetic certificate

The following Python script uses only exact rational arithmetic. It constructs the distribution, verifies the marginal and negative-correlation conditions by exhaustive enumeration, and then computes the optimal value of the weighted random-order online selector by a full-history Bellman recursion. The state records, for each of the five elements, whether it is unseen, seen inactive, or seen active and rejected. Zero-probability branches are not expanded.

```
from fractions import Fraction
from functools import lru_cache
from itertools import product

# Parameters.
a = Fraction(1, 100)
p = Fraction(99, 400)
w = (Fraction(43), Fraction(1), Fraction(1), Fraction(1), Fraction(1))
N = 5

# Activation distribution on vectors (Z_0,...,Z_4).
prob = {}

def add(vec, pr):
    prob[tuple(vec)] = prob.get(tuple(vec), Fraction(0)) + pr

# Branch Z_0=1.
add((1,0,0,0,0), a*a)
for i in range(1,5):
    vec = [1,0,0,0,0]
    vec[i] = 1
    add(vec, a*p)

# Branch Z_0=0.
for bits in product([0,1], repeat=4):
    pr = 1 - a
    for bit in bits:
        pr *= p if bit else 1-p
    add((0,) + bits, pr)

assert sum(prob.values()) == 1

# Marginals.
z = [sum(pr for vec, pr in prob.items() if vec[i] == 1) for i in range(N)]
assert z[0] == a
for i in range(1,5):
    assert z[i] == p
assert sum(z) == 1

# Exhaustive check of setwise negative correlation.
for b in [0,1]:
    for mask in range(1 << N):
        S = [i for i in range(N) if (mask >> i) & 1]
        lhs = sum(pr for vec, pr in prob.items()
                    if all(vec[i] == b for i in S))
        rhs = Fraction(1)
        for i in S:
            rhs *= z[i] if b == 1 else 1-z[i]
        assert lhs <= rhs, (b, S, lhs, rhs)
```

```

# Full-history dynamic program.
# obs[i] = -1 means unseen, 0 means seen inactive,
# and 1 means seen active and rejected.
def consistent(vec, obs):
    return all(obs[i] == -1 or obs[i] == vec[i] for i in range(N))

@lru_cache(None)
def V(obs):
    obs = tuple(obs)
    unseen = [i for i, status in enumerate(obs) if status == -1]
    if not unseen:
        return Fraction(0)

    denom = sum(pr for vec, pr in prob.items() if consistent(vec, obs))
    assert denom > 0

    total = Fraction(0)
    for i in unseen:
        active_num = sum(pr for vec, pr in prob.items()
                        if consistent(vec, obs) and vec[i] == 1)
        q = active_num / denom

        contribution = Fraction(0)
        if q > 0:
            obs_active = list(obs)
            obs_active[i] = 1
            contribution += q * max(w[i], V(tuple(obs_active)))
        if q < 1:
            obs_inactive = list(obs)
            obs_inactive[i] = 0
            contribution += (1-q) * V(tuple(obs_inactive))

        # Conditional on the current history, the next unseen element
        # is uniformly distributed over the unseen elements.
        total += contribution

    return total / len(unseen)

opt = V(tuple([-1]*N))
assert opt == Fraction(2290330421101, 2560000000000)
assert opt < Fraction(179, 200)
assert Fraction(179, 200) < Fraction(79, 125) * Fraction(71, 50)
assert opt / Fraction(71, 50) == Fraction(2290330421101, 3635200000000)

print(opt)
print(float(opt))
print(opt / Fraction(71, 50))

```

## References

- [1] Brian Brubach, Nathaniel Grammel, Will Ma, Calum MacRury, and Aravind Srinivasan. *Improved Guarantees for Offline Stochastic Matching via New Ordered Contention Resolution Schemes*. arXiv:2106.06892v3, 2026.
- [2] Eric Fithian, Rad Niazadeh, and Pranav Nuti, maintainers. *Open Problems in Operations Research*, problem W3171990983\_p0. [Problem page](#).